Research Statement

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I am a behavioral economist studying how boundedly rational behavior shapes competition and coordination. I combine game theory and laboratory experiments to examine how people acquire and process information and reason about others under different information structures and strategic environments. My goal is to use this understanding to inform the design of market mechanisms.

1. Consumer Search and the Market for Prominence

My job market paper, "Search Frictions and Welfare in a Market for Prominence: Theory and Experiment," studies how consumer search costs affect the level and distribution of surplus when sellers compete for prominence. In online marketplaces, consumers can often obtain price information with little effort, but learning whether a product suits their tastes or needs remains costly. Consumers may stop before inspecting every product, giving sellers a reason to pay for search ads and sponsored listings that make their products easier to discover and inspect.

I develop a model of the market for prominence in which two *ex ante* identical sellers bid for a prominent position in a second-price auction and then simultaneously post prices. The buyer observes both prices and learns her match value for the prominent seller's product for free. She then decides whether to pay a search cost to inspect the other, non-prominent seller's product. Because prices are visible, sellers can use them strategically to influence whether the buyer searches further. I use a between-subjects laboratory experiment with low and high search-cost treatments to test the model's predictions for welfare and behavior.

In the laboratory, I find systematic departures from the predicted distribution of surplus, which I explain using observed bidding, pricing, and search behavior. Platform auction revenue changes little as search cost rises, despite the predicted increase, reflecting the weak response of the losing bids that determine auction payments. Consumer surplus exceeds equilibrium predictions in both treatments and falls proportionally less than predicted as search cost rises. Prominent sellers charge lower prices than predicted, benefiting buyers. These lower prices also encourage buyers with a good initial match to buy from prominent sellers without further search, helping them save the cost of another inspection. Sellers, however, jointly retain around 20 percent of total surplus after paying for prominence. This is because they earn less revenue than predicted and pay a substantial share of it to the platform. The same low prices that benefit buyers reduce sellers' revenue per sale. Sellers also lose sales when high prices at non-prominent sellers discourage further search by buyers who find the prominent product a bad match. Some of these buyers exit without purchasing from either seller.

2. Depth of Strategic Reasoning

My research on bounded rationality examines where people start when reasoning about others and how far they proceed. In joint work, "Revisiting the Intra-Team Communication Method to Elicit Level-k Reasoning in Beauty Contests and 11–20 Games," we provide the first replication of Burchardi and Penczynski's beauty-contest experiment and extend their method to variants of the 11–20 game. Participants explain their choices in incentivized messages to teammates. We combine these messages with their choices to infer what they believe a non-strategic player would do and how

many steps of reasoning they take from that starting point.

In the beauty contest, beliefs about non-strategic play are consistent with the observed behavior of players classified as non-strategic. This belief-action consistency breaks down in the reordered 11–20 game: observed non-strategic choices are more dispersed than higher-level players expect, and both choices and beliefs depart from the standard assumption about non-strategic play. Using our within-subject design, we also find that reasoning levels inferred from messages are positively correlated across games, a relationship not revealed by choices alone. This provides tentative support for viewing depth of reasoning as a stable individual characteristic.

3. Communication and Coordination

In "Coordination under Uncertainty and Noisy Communication: A Generalized Email Game," I study how communication rules affect coordination under asymmetric information and uncertain signal delivery. In the model, one player knows whether the state favors a risky joint action, while the other does not. A single signal can reveal the favorable state to the uninformed player but has a fixed probability of failing to arrive. The model permits both an inefficient equilibrium in which the signal is ignored and a more efficient one in which players act on it, making equilibrium selection the central empirical question. In a between-subjects laboratory experiment, I vary the communication protocol across three treatments: automatic sending, free voluntary sending, and costly voluntary sending.

In the laboratory, I find that free voluntary sending by the informed player makes players more likely to take the risky action after favorable information than automatic sending does. Imposing a message cost on the informed player, however, discourages sending and thereby reduces coordination. These results suggest that choosing to send can convey an intention to coordinate when interests are aligned, even without common knowledge of the state. A fee intended to enhance signal credibility can nevertheless backfire by discouraging communication.

4. Research Pipeline

I plan to extend my work on consumer search in two directions, both theoretically and experimentally. First, I will examine a platform's incentives to make search easier or harder when it earns revenue from advertising and other sources, such as sales commissions. I will compare its preferred search cost with the socially optimal level to assess when regulation could improve welfare. Second, I will compare the efficiency of alternative mechanisms for allocating prominence, including different auction formats, when position values depend on subsequent pricing and consumer search.

In the long run, I will bring my work on strategic reasoning and communication into market settings to study how rules governing allocation and information conveyance affect efficiency and the distribution of surplus. In laboratory comparisons of these rules, I will use participants' choices and reasoning to explain why certain mechanisms perform better than others.